

Week 4: Project Setup and Backend Integration

Overview:

Week 4 focused on initiating actual development, starting with backend integration and core project configuration. Reliable cloud connectivity and authentication are crucial for real-time location-based systems.

Progress:

- Created a new Android Studio project and set up build configurations.
- Integrated Firebase into the project and verified app registration on Firebase Console.
- Configured Firebase Authentication for secure user login and registration.
- Designed Firestore data structure to store station metadata and facility coordinates.
- Enabled Firestore indexing and optimized read/write operations.
- Set Firebase rules for access control and ensured basic data security.
- Built and tested login and signup screens connected to Firebase backend.
- Created test documents in Firestore and verified data fetch functionality.
- Documented project structure and backend logic for future reference.

Summary:

The application's backend layer is functional with Firebase Authentication and Firestore fully integrated. Secure login and structured location data storage form the backbone of our navigation system.

Next Week and Beyond:

The focus will now shift to ARCore integration, environmental scanning, and real-time marker placement in the app interface.